

AI-Powered Product Matching System

Architecture — Snowflake Cortex Hackathon Submission

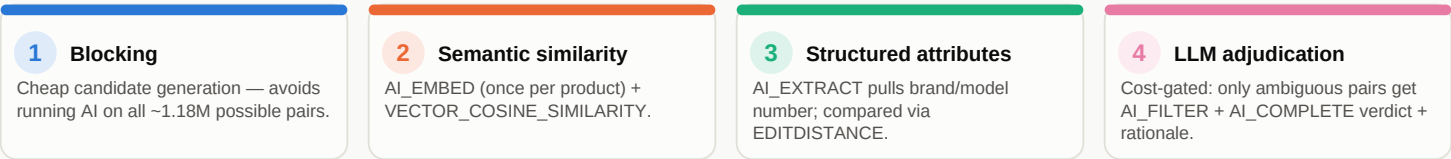
Cross-retailer entity resolution • 4-signal ensemble matching • 3 Cortex Agents • MCP-integrated

Problem & Approach

Retailers need to know which of their products a competitor also sells, and at what price, to compete on pricing. This system resolves that as an **entity-resolution problem**: given two independent product catalogs (Abt.csv, Buy.csv — the standard Abt-Buy benchmark, 1,081 and 1,092 products, no shared IDs) with no clean join key, determine which pairs refer to the same physical product, then layer competitive pricing analysis on top.

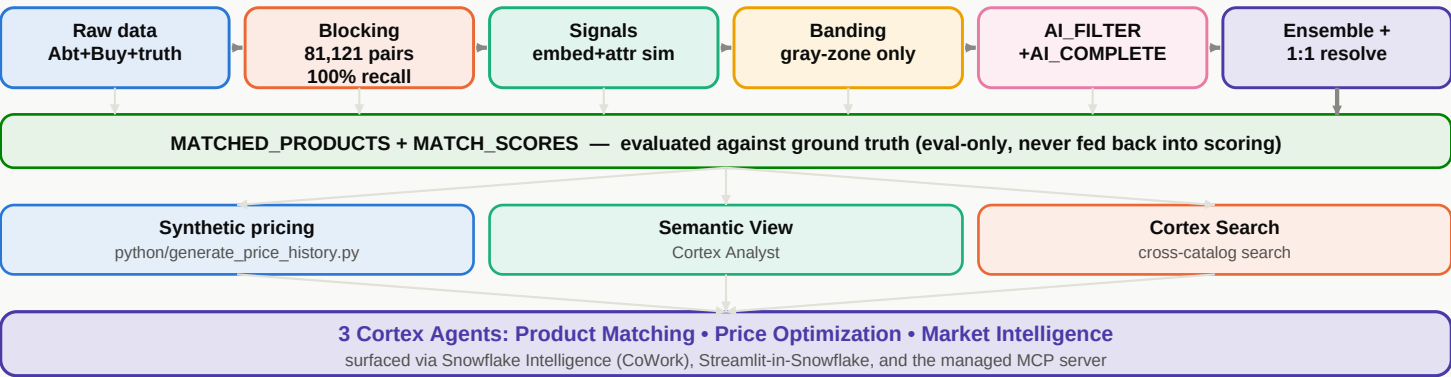
Differentiator: multi-strategy ensemble matching

Most naive approaches run one embedding-similarity pass and threshold it. This system instead combines four independent signals into one explainable score **and rationale** per candidate pair:

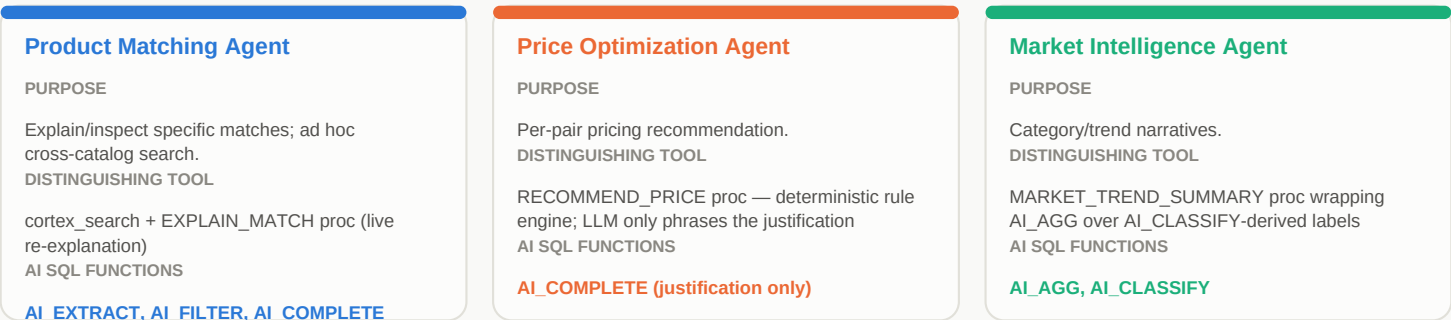


A weighted combination produces final_confidence (MATCH >= 0.70 / REVIEW 0.40–0.70 / NO_MATCH otherwise), followed by a greedy 1:1 conflict-resolution pass (Python stored proc) — no product is claimed twice.

Pipeline



Three Agents, Three Distinct Techniques



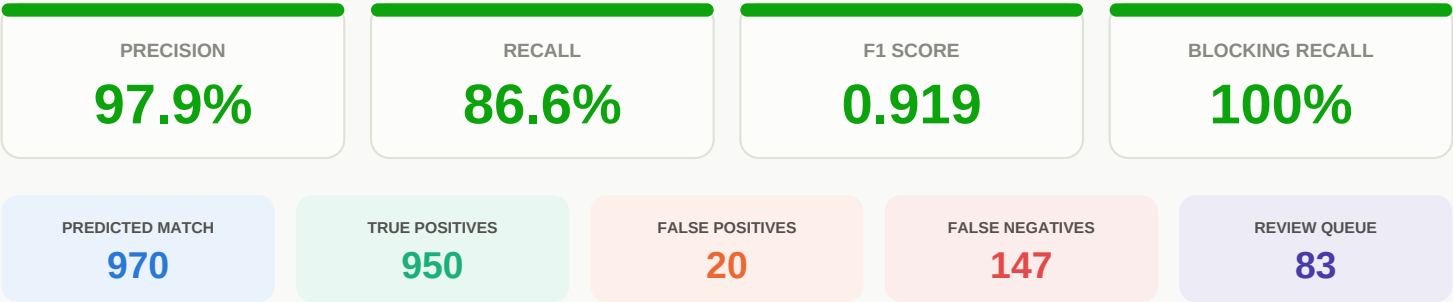
All three sit on one semantic view (ABT_BUY_SEMANTIC_VIEW) so the same objects back the Cortex Agents, the Snowflake Intelligence/CoWork chat surface, the Streamlit dashboards, and the MCP server — no duplicated build effort.

Synthetic Data Disclosure

 **The Abt-Buy dataset has no time-series pricing — pricing history here is simulated**

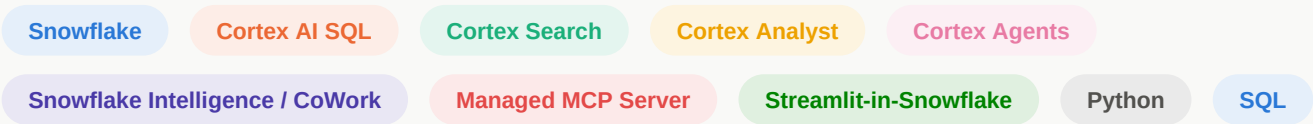
Only a single point-in-time price exists per product, missing on 46–62% of rows. `generate_price_history.py` fabricates an 8–12-week weekly series per matched pair per retailer (real price where available, else bootstrapped, plus random-walk drift and occasional promo/undercut events). Every row is tagged `is_synthetic = TRUE` end-to-end and shown as a persistent banner in the dashboard. **Product identities and matches are computed entirely from the real dataset** — only the price history is simulated.

Results



Measured live via `V_MATCHING_ACCURACY` against the real `abt_buy_perfectMapping.csv` ground truth. Precision is intentionally the priority — a `MATCH` label is right 98% of the time. Ambiguous pairs land in a separate **REVIEW** tier rather than being forced into `MATCH` or `NO_MATCH`, feeding a human-in-the-loop review queue instead of a false claim of certainty.

Tech Stack



Every match is explainable. Every price is justified. Every trend is narrated.

Built entirely on Snowflake Cortex — AI SQL, Search, Analyst, Agents, and a managed MCP server.